

Information and Opinion in Social Network

Degroot Model

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- Your opinion, knowledge, beliefs are influenced by people and media around you.
- We can model such situation as a network of people and information source, where each individual learns from and is influenced by others through his/her connection to them.
- The dynamics of information and opinions may be shaped by the structure of the underlying social network.

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We may be interested in questions such as

- How are opinions/beliefs formed in social information network?
- How do they spread?
- How do people reach a consensus/how do people come to have opposing opinions/beliefs?
- Does the underlying network structure matter for all of these?
- Where did such a network come from? How does it change over time?
- Where and how should we intervene?

Degroot Model

- Degroot model (1971) is the simplest kind of such network model. It can be regarded as a directed network augmented with a number (opinion, belief etc.) assigned to each node and a weight assigned to each link (which represents the importance of the neighbor's opinion).
- It is a very simple model that generates a tractable dynamics of opinions. It is also useful to study consensus.
- It does not capture many important aspects of real information network (too simplistic dynamics, for example). But it can be a useful starting point/benchmark for analysis.

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- People $i = 1, \dots, n$.
- i 's belief/opinion $x_i \in [-1, 1]$ (normalized).
- Updating dynamics: $x_i(t+1) = \sum_j T_{ij} x_j(t)$, initial $\mathbf{x}(0)$ is given, where $T_{ij} \geq 0$ is the weight of j 's opinion for i and satisfies $\sum_j T_{ij} = 1$.

It is convenient to express this dynamics with **right stochastic matrix** T (i.e. $\sum_j T_{ij} = 1$):

$$\mathbf{x}(t+1) = T\mathbf{x}(t)$$

where $\mathbf{x}(t) = (x_1(t), \dots, x_n(t))^T$ is a column vector.

Remark.

- This matrix T can be regarded as an adjacent matrix for a generalized directed graph with weights. T_{ij} can be expressed as an arrow from j to i with positive weight (instead of just 0 or 1).
- However, T_{ii} can be naturally positive in this context, as it represents a weight on i 's own opinion as he/she updates his/her opinion.
- Is this repeated updating rule reasonable?
 - ▶ A change of your neighbor's opinion suggests some new information.
 - ▶ On the other hand, such a change may be partly caused by your influence on the neighbor. So, it is not clear how much of the change reflects a new information.

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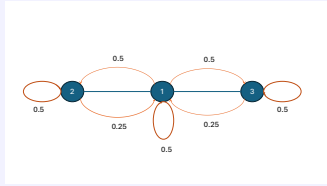
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We consider a few examples.

Example 1.



- $n = 3$.
- 1 (influencer) gives 50% weight to its own opinion and 25% to the opinion of 2 and 3 (i.e. $T_{11} = 1/2$, $T_{12} = 1/4$, $T_{13} = 1/4$)
- 2 and 3 give 50% weight to its own opinion and 50% to the influencer.

- Suppose that $x_1(0) = 1$ and $x_2(0) = x_3(0) = -1$. Initially 1 and 2, 3 have the opposite opinions. But $x_1(1) = x_2(1) = x_3(1) = 0$, hence $x_1(t) = x_2(t) = x_3(t) = 0$ for any t after that. So, they reach a consensus immediately.
- The updating formula is

$$x_1(t+1) = 1/2x_1(t) + 1/4(x_2(t) + x_3(t))$$

$$x_z(t+1) = 1/2x_1(t) + 1/2x_z(t), \quad z = 2, 3$$

This can be simplified if we introduce $\bar{x}_{23}(t) = 1/2x_2(t) + 1/2x_3(t)$:

$$x_1(t+1) = 1/2x_1(t) + 1/2\bar{x}_{23}(t)$$

$$\bar{x}_{23}(t+1) = 1/2x_1(t) + 1/2\bar{x}_{23}(t)$$

Hence $x_1(t) = \bar{x}_{23}(t)$ for any $t \geq 2$ in general for this example.

- Let's try another starting point: $x_1(0) = 1$, $x_2(0) = -1/3$, and $x_3(0) = -1$.
- Since $\bar{x}_{23}(0) = -2/3$, $x_1(t) = \bar{x}_{23}(t) = 1/6$ for $t \geq 1$ by the previous observation.
- Note that $x_2(t) > 1/6 > x_3(t)$ at each t , but they are both converging to $1/6$ as $t \rightarrow \infty$.

Example 2.

- Ring network with 6 nodes $\{1, 2, 3, 4, 5, 6\}$.
- Assume that i 's opinion is updated to the average opinion of i 's two neighbors at each step (i.e. $T_{ii-1} = T_{ii+1} = 1/2$).
- Pick $x_i(0) = 1$ for even i and $x_i(0) = -1$ for odd i as a starting point. How does $x(t)$ change? Do they reach a consensus?
- What if there are 5 nodes?

Consensus

- People reach a consensus in the first example. The second example suggests that it is not necessarily the case.
- When does a consensus emerge?
 - ▶ It is clear that there cannot be a group of people who are not influenced by the rest.
 - ▶ Even if everyone influences everyone indirectly (i.e. the underlying directed graph of influence is strongly connected), the updating process may still lead to some cycle.

It turns out that a slightly stronger notion of connectedness is sufficient for a consensus.

Suppose that there is t such that T^t is a positive matrix ($T_{ij}(t) > 0$ for any i, j). This means that everyone is influenced by everyone indirectly in t steps. This condition is satisfied when the network of influence is strongly connected and there is at least some i with $T_{ii} > 0$.

Then, for any initial set of opinions $x(0)$, the society reaches a consensus eventually, i.e. there is $x^* \in [-1, 1]$ such that $x_i(t) \rightarrow x^*$ for every i .

This follows from the following useful fact about stochastic matrix.

Theorem

Suppose that T is a right stochastic $n \times n$ matrix and there is a positive integer t such that T^t is a positive matrix. Then T^t converges to some right stochastic matrix T^* with identical row $w^* = (w_1^*, \dots, w_n^*)$ as $t \rightarrow \infty$.

Hence $x_i(t)$ converges to $w^* \cdot x(0)$ for every i .

This is a well-known fact for **Markov chains**. Consider a stochastic process where T_{ij} is the probability that the state changes from i to j . Pick the initial state randomly according to distribution $p(0) = (p_1(0), \dots, p_n(0))$. Then the state distribution in period t is $p(t) = p(0)T^t$. So, the above theorem means that the long-run distribution $p(\infty) = \lim_{t \rightarrow \infty} p(0)T^t = p(0)T^* = w^*$ is independent of the $p(0)$ (assuming T^t is positive for some t).

- What is this w^* ?
- We can get an approximate value from T^t for large t .
- Or note that, for any distribution $p = (p_1, \dots, p_n)$, we have $w^* = pT^* = p(\lim_{t \rightarrow \infty} T^t)$. Hence $w^* = (p \lim_{t \rightarrow \infty} T^{t-1})T = w^*T$. So, we can find w^* by solving $w^* = w^*T$ (Mathematically, w^* is a left eigenvector of T with eigenvalue 1).
- Let's go back to Example 2 with 5 nodes. The dynamic process of opinions is quite complicated. But it is easy to see that $w^* = (1/5, \dots, 1/5)$, so we can immediately see that their opinions converge to $1/5$.

Influence

- The previous result shows that everyone's opinion converges to $w^* \cdot x(0) = \sum_i w_i^* x_i(0)$ in the long run, which is a weighted average of initial opinions.
- $w_i^* \in (0, 1)$ measures the influence of i 's initial opinion over the long-run consensus. The larger w_i^* is, the more influential i is.

Wisdom of Crowd

Suppose that everyone is equipped with a piece of information about the truth, and interpret the above dynamics as a process to aggregate information in the society.

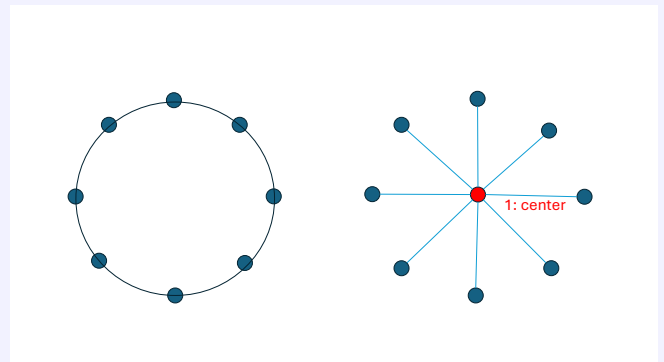
More formally, we consider the following setting:

- $\theta \in \mathbb{R}$: true state
- $x_i(0) = \theta + \epsilon_i$, where ϵ_i is i.i.d. noise with mean 0 ($E[\epsilon_i|\theta] = 0$). So each person observes an independent and unbiased signal about θ .
- Network with n nodes and T . Assume that T^t is a positive matrix for some t

We know that $\lim_{t \rightarrow \infty} x_i(t) = x(\infty)$ for every i . Does this $x(\infty)$ reveal the true state θ for a large society?

Which is better?

- If we can observe all $x_i(0)$ to estimate θ , then our best estimate of θ is $\frac{\sum_{i=1}^n x_i(0)}{n}$. Law of large number tells us that this converges to θ with probability 1 as $n \rightarrow \infty$.
- If the underlying directed network is the clique with self-link and equal weights (i.e. $T_{ij} = \frac{1}{n}$ for all i, j), then $x(\infty)$ coincides with the above best estimate for any n .
- How good would $x(\infty)$ be for other networks?



Ring: $T_{i-1} = T_{ii} = T_{i+1} = 1/3$. Star: $T_{1j} = 1/n$ for every j , $T_{ij} = 1/2$ for $i \neq 1, j = 1, i$.

- For the ring, $w_i^* = \frac{1}{n}$, hence $x(\infty) = \frac{\sum_{i=1}^n x_i(0)}{n}$.
- For the star, solve the following equations:

$$\frac{1}{n}w_c^* + 0.5(n-1)w_p^* = w_c^*$$

$$\frac{1}{n}w_c^* + 0.5w_p^* = w_p^*$$

$$w_c^* + (n-1)w_p^* = 1$$

Then we have $w_c^* = \frac{n}{3n-2}$ (for center) and $w_p^* = \frac{2}{3n-2}$ (for peripherals).

- In general, the necessary and sufficient condition for $x(\infty)$ to converge to θ as $n \rightarrow \infty$ for a sequence of networks (that generate a consensus) is that w_i^* converges to 0 uniformly across nodes as $n \rightarrow \infty$.
- This condition is satisfied for ring networks, but not for star networks.
- **Exercise:** Can you adjust the weights in the sequence of star networks to achieve this property? How?

Summary

- Degroot model: a simple model of dynamic opinion formation and information aggregation.
- Strong connectivity + α for T implies that the society achieves a consensus.
- Each node's influence over the consensus is obtained from the left eigenvector of T with eigenvalue 1.
- As a model of information aggregation, the consensus converges to the true state as $n \rightarrow \infty$ if and only if the influence weight of every node converges to 0 uniformly as $n \rightarrow \infty$.

Suggested reading. Golub and Jackson (2010)